

- **Homework Set 1. (AI Introduction)**
- **3 problems. Due Oct 10, 2026**
- **Each problem is 10 points.**
- **Homework: INTRO1**
- “act rationally”
 - What does the word “rationally” means?

Hint: (a) In economics most theories assume players/humans are selfish, but act **rationally**.

(b) You may use Chinese words here 用中文表达 if that's easier.

Healthy Perspective of AI

- The purpose of AI is to improve/innovate technology
- Many tasks, humans' way is good, provide inspiration.
- Some tasks humans don't do well:
 - Compare two pictures of the same person to check if they are some person
- The over-all drive is to develop new/effective technology to solve more tasks, not necessarily blindly follow human's way!

Homework Intro2:

Can you think of a task that humans don't do well.

Homework Intro3:

From 2-layer network to multi-layer network

- In 1970, neural network is dead because it could not solve the exclusive-or problem (one of the four basic computer operations)
- Many years later multi-layer network emerges which can easily solve the X-OR problem.
- Ask this problem on your favorite LLM, on specific time, why, the new innovations occurred in-between.
- Adding one layer in the network seems as simple as $1+2=3$ as most people would think. Is it really? This task takes humans how many years to accomplish?